

RAJ S

Nagarajan Shunmugam

AI Engineer · CRISPR & NGS Researcher · Co-Founder & CTO · Future Veterinarian (DVM Candidate)

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PROFILE

Nine years of applied AI across six countries: autonomous trucks at Scania, medical imaging in Berlin, AI for the world's largest veterinary community, and CRISPR machine learning in Wrocław. Three-time startup co-founder. Currently a CRISPR & NGS data researcher at the Hirsfeld Institute (Polish Academy of Sciences), CTO of MilkAI, and in the final year of veterinary school (graduating December 2026), building the bridge between AI and animal health.

EXPERIENCE

- CRISPR & NGS Data Researcher** · Hirsfeld Institute of Immunology & Experimental Therapy, PAN
Wrocław, Poland
Nov 2025 – present
ML pipelines for CRISPR off-target prediction on genomic, NGS and multi-omics data (k-mer, scalar, one-hot encodings; CNN and Transformer models). Multi-agent AI systems that design and validate wet-lab experiment protocols.
- Co-Founder & CTO** · MilkAI LLC
California, USA
Jul 2025 – present
Product studio shipping production-ready web, mobile and AI-first apps, with design-to-deploy CI/CD. Dozens of launches.
- Artificial Intelligence Engineer** · Veterinary Information Network
Remote (US company)
May 2023 – Jul 2025
AI engineering for the world's largest online veterinary community.
- Co-Founder** · Zaide.ai
Remote
Jan 2023 – Aug 2024
Built private, custom large language models for healthcare and life sciences: secure local LLMs designed so sensitive clinical and research data never leaves the organization.
- Mentor & Alpha Tester** · Coursera
Orange County, CA
Apr 2017 – Jun 2023
Six years mentoring the Neural Networks & Deep Learning course; alpha-tested unreleased AI courses; mentored 3D-printing applications.
- Assistant Lecturer** · Coventry University (Wrocław campus)
Wrocław, Poland
Aug 2022 – Mar 2023
Taught AI, IoT and software engineering.
- Senior AI Instructor** · Techeta Technologies
India
Sep 2021 – Sep 2022
Trained technical and corporate teams in applied AI.
- Co-Founder & CTO** · Neurons4vet.ai
Toronto, Canada
Jan 2020 – Dec 2021
First venture at the intersection of AI and veterinary medicine.
- Cohort Member** · Entrepreneur First
Berlin, Germany
Mar – Aug 2021
Europe's leading deep-tech talent investor (300+ startups, over \$2.7B portfolio).
- Thesis Student** · Scania Group
Stockholm, Sweden
Jan 2020 – Mar 2021
Truck visual perception: classification, detection, semantic segmentation, depth estimation, reinforcement learning.
- Machine Learning Engineer** · ai4medicine
May – Oct 2020

Berlin, Germany

Computer-vision AI for stroke and bleeding detection.

Senior Design Engineer · KTH Formula Student

Aug 2019 – Jan 2020

Stockholm, Sweden

Autonomous race car perception; LiDAR point clouds; NVIDIA Jetson deployment.

Project Assistant · Indian Institute of Technology, Delhi

Sep 2017 – Feb 2018

New Delhi, India

Driverless car: PID tuning, GPS+IMU+LiDAR+radar sensor fusion.

Research Scholar · Indian Institute of Technology, Madras

Mar – Aug 2017

Chennai, India

Embedded programming, biomedical research lab (bio-medical kiosks, SpO2 sensors).

EDUCATION

Wrocław University of Environmental and Life Sciences (UPWR) · Veterinary

2021 – 2026

Medicine, Veterinary Surgeon (DVM), final year, graduating December 2026

Technische Universität Berlin · M.Sc. Autonomous Systems, Robotics & AI (full scholarship)

2018 – 2021

KTH Royal Institute of Technology · M.Sc. Autonomous Systems, Robotics & AI (EIT Digital double degree)

2018 – 2021

Aalto University · EIT Digital Summer School: Disrupting Retail

2019

Anna University · B.E. Electrical & Electronics Engineering (partial scholarship)

2013 – 2017

SKILLS

AI/ML: CNNs & Transformers, deep reinforcement learning, imitation learning, multi-agent systems, data analysis ·

Genomics: CRISPR guide design, NGS & multi-omics, k-mer/one-hot encodings · **Autonomy:** detection & segmentation, depth estimation, LiDAR, sensor fusion, NVIDIA Jetson · **Engineering:** technical leadership, IoT & embedded (Arduino), 3D printing, CI/CD

PUBLICATION & HONORS

Publication: Deep Learning for Hardware-Constrained Cars: Imitation Learning & Deep Reinforcement Learning

Honors: Full scholarship (master's) · Partial scholarship (bachelor's) · **Languages:** Bilingual, English (IELTS C1) and Tamil